

DiagAgent: An Agent Framework for Power Equipment Fault Diagnosis by Integrating RAG and MCP tools

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Abstract—Large language models (LLMs) show promise for diagnosing power equipment faults. However, they face issues such as hallucinations, knowledge lag, and a lack of specialized analytical capabilities. This paper introduces DiagAgent, an innovative Artificial Intelligence (AI) agent framework. It synergizes Retrieval-Augmented Generation (RAG) with domain-specific small models through the Model Context Protocol (MCP). The framework first constructs a knowledge base containing equipment information and historical cases. Next, it employs a multi-level hybrid retrieval strategy to obtain relevant information as the intelligent core. The LLM then dynamically invokes specialized small-model tools encapsulated via MCP, which include analyzers for operating data, SF₆ pressure, and simulations, to conduct in-depth analysis. Finally, it generates structured diagnostic reports. A power bushing diagnosis case, implemented on the Dify platform, demonstrates that DiagAgent significantly outperforms baseline methods, achieving a diagnostic accuracy of 91.3% and a report quality score of 4.4/5. This work provides a practical and scalable blueprint for developing reliable, domain-specific AI assistants in the power industry and beyond.

Keywords—fault diagnosis; large language models; retrieval-augmented generation; model context protocol; AI agent; power equipment

I. INTRODUCTION

The reliability of critical power equipment like bushings is fundamental to grid security [1]. Traditional diagnostic methods, ranging from expert systems [2] to data-driven deep learning models [3], often struggle with limitations in scalability, knowledge maintenance, and interpretability. The

advent of Large Language Models (LLMs) offers a transformative approach with their advanced natural language understanding and reasoning capabilities [4]. However, applying general-purpose LLMs directly to the high-stakes domain of power equipment diagnosis introduces critical risks: (1) Hallucination, leading to incorrect conclusions [5]; (2) Static knowledge, unable to access the latest device information and case studies; and (3) Lack of analytical depth, incapable of executing domain-specific numerical computations.

For instance, when diagnosing a bushing with insulation material introduced after 2023, a GPT-4-based system may suffer from knowledge lag, providing recommendations based on outdated standards. More critically, it can hallucinate non-existent failure modes. Our internal tests indicate that such fabricated content accounts for over 15% of its suggestions in edge cases, posing direct operational risks.

Retrieval-Augmented Generation (RAG) effectively mitigates these issues by grounding LLMs in external, authoritative knowledge bases [6]. However, standard RAG lacks the capability for deep, signal-based analysis. In parallel, the Model Context Protocol (MCP) has emerged as an open standard for seamless tool integration, enabling LLMs to leverage external computational resources.

To bridge this gap, we propose DiagAgent, an intelligent agent framework that integrates the knowledge-awareness of RAG with the analytical power of specialized small models via MCP. The core innovation is not merely a toolchain but a synergistic system where components compensates for each other's weaknesses. This enables the agent to retrieve context-

aware knowledge and perform specialized computations on demand, a capability critical to trustworthy industrial diagnosis.

Our contributions are threefold:

1. Propose an intelligent agent framework for fault diagnosis that integrates RAG with MCP tool invocation to address both knowledge and analytical challenges.
2. Design a multi-level hybrid retrieval strategy for fault diagnosis to enhance knowledge retrieval accuracy.
3. Implement the prototype system on the Dify platform and validate its effectiveness using real-world bushing failure data.

II. RELATED WORK

A. LLMs for Industrial Diagnostics

LLMs have shown initial promise in industrial applications, primarily serving as static question-answering systems built on manually curated knowledge bases [7]. A critical limitation of this paradigm is its lack of autonomous tool-use and dynamic analytical capabilities—precisely those required to process and reason over multi-source data (e.g., technical manuals, sensor logs, and historical cases) in complex diagnostic scenarios. This gap motivates the need for an agent-centric framework capable of actively retrieving relevant information and invoking specialized tools for deeper analysis.

B. Retrieval-Augmented Generation

RAG significantly enhances the factual accuracy and knowledge currency of LLMs by dynamically retrieving relevant information from external sources during the generation process [7]. Advanced RAG methodologies [6], such as query rewriting and hierarchical retrieval, have further optimized this paradigm, leading to notable improvements in open-domain question answering.

However, these advancements are primarily designed for and evaluated on unstructured textual corpora. A key limitation emerges when applied to complex industrial domains like power systems, where diagnostic knowledge is inherently distributed across heterogeneous data formats, including both unstructured documents and structured records [8]. This

discrepancy highlights a gap in existing retrieval strategies for specialized, multi-modal knowledge fusion.

C. Model Context Protocol

The capacity of LLMs to utilize tools marks a pivotal step toward building more capable AI systems. While frameworks such as ReAct [9] have explored this direction, the MCP further establishes a standardized, vendor-agnostic protocol for tool integration, significantly enhancing flexibility and interoperability in the development of complex intelligent agents. As an open protocol, MCP uniformly governs interactions between LLMs and external tools or data sources. Compared to vendor-specific solutions, its openness and generality make it better suited for supporting the development and integration of sophisticated agent systems [10].

D. Dify platform

LLM application development platforms provide the foundational infrastructure for building and deploying AI applications. They integrate key tools, including visual workflow orchestration, RAG, and agent design, enabling rapid development of diagnostic agents that combine domain knowledge with analytical tools for power equipment fault diagnosis.

Among mainstream platforms, LangChain emphasizes programming flexibility, LlamaIndex specializes in optimizing RAG processes, and Coze is geared towards quickly building lightweight conversational agents [6]. In contrast, Dify is an open-source, full-stack LLM operations platform. It provides distinct advantages in visual orchestration, enterprise integration, and production readiness, directly enabling the flexible design of complex diagnostic workflows. Additionally, its robust API gateway, fine-grained access control, and mature solutions for private deployment ensure that systems based on the DiagAgent framework can achieve high diagnostic performance while meeting the core industrial requirements for stability, security, and maintainability.

III. THE DIAGAGENT FRAMEWORK

The architecture of DiagAgent, depicted in Figure 1, is implemented on the Dify platform and consists of four core components: a Knowledge Base, a Retrieval Module, an LLM-based Agent with an MCP Toolbox, and a Report Generator.

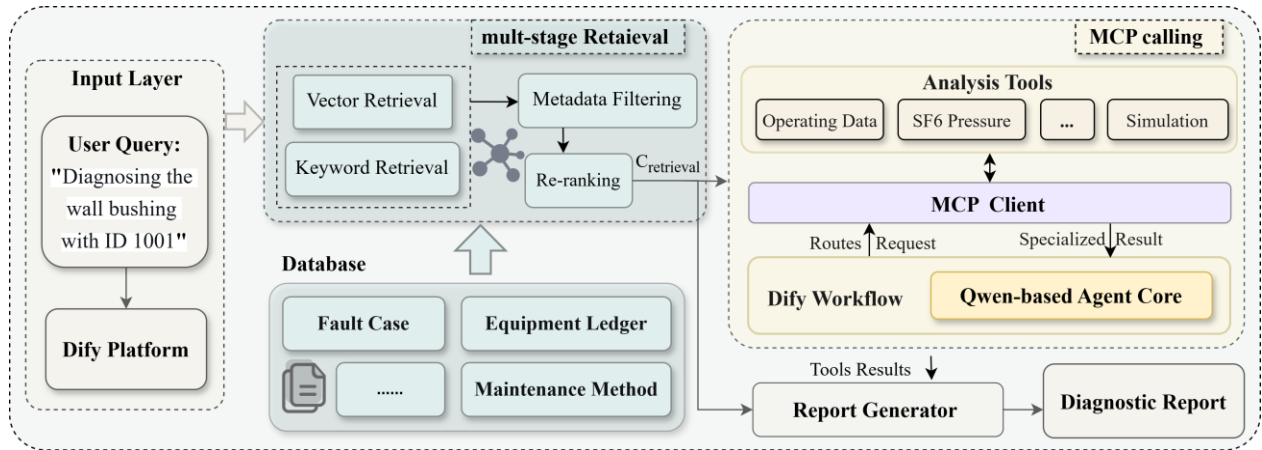


Figure 1. System Architecture of DiagAgent

A. Knowledge Base and Multi-Stage Retrieval

The framework's knowledge base integrates heterogeneous data sources, including semi-structured data (e.g., equipment parameters and operational procedures) and unstructured texts (e.g., historical fault records). This raw data is processed through a cleaning, text segmentation, and vectorization pipeline before being stored in a vector database.

Given a query q , the objective is to retrieve the most relevant context $C_{retrieved}$ from the knowledge base. To this end, we employ a multi-level hybrid retrieval strategy, which operates through the following key stages:

1) *Vector Retrieval*: A dense passage retrieval (DPR) model fetches semantically relevant chunks. Initial screening based on semantic vector similarity, calculated as (1):

$$Sim_{vec}(q, d) = \frac{E_{(q)} \cdot E_{(d)}}{\|E_{(q)}\| \|E_{(d)}\|} \quad (1)$$

where $E_{(*)}$ denotes the text embedding model. The N segments with the highest similarity form the set D_{vec} .

2) *Keyword Retrieval*: The BM25 algorithm performs sparse retrieval for exact keyword matches, as in (2):

$$Score_{BM25}(q, d) = \sum_{i=1}^n IDF(t_i) \cdot \frac{f(t_i, d) \cdot (k_1 + 1)}{f(t_i, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{avgdl})} \quad (2)$$

The M segments with the highest scores form the set $D_{keyword}$.

3) *Metadata Filtering*: Results are filtered by metadata such as equipment ID.

4) *Re-ranking*: Reorder the candidate set $D_{candidate} = D_{vec} \cup D_{keyword}$. The final score is a weighted combination, as in (3):

$$Score_{final}(d) = \alpha \cdot Sim_{vec}(q, d) + \beta \cdot Norm(Score_{BM25}(q, d)) \quad (3)$$

The M segments with the highest scores form the set $D_{keyword}$. where the weight coefficients α and β , set to 0.7 and 0.3 via grid search, balance semantic recall against keyword precision. Select the top K fragments with the highest scores as the final retrieval context $C_{retrieved}$.

B. LLM-based Agent and MCP-Based Tool Calling

Specialized small models are wrapped as tools on an MCP server, as depicted in Figure 2. In this work, Qwen-30B [11] functions as an autonomous agent acting as an autonomous agent, decides based on the query and context whether to invoke a tool, which one to use, and with what parameters. This creates a powerful synergy where the LLM handles high-level reasoning and the small models provide precise, domain-specific computations.

The LLM agent decides whether to invoke a tool based on the query q and the context $C_{retrieved}$. It first evaluates information sufficiency:

$$S_{sufficiency} = LLM_{judge}(q, C_{retrieved}) \quad (4)$$

If $S_{sufficiency} < \tau$ (where τ is the threshold), the tool must be invoked. The agent selects the most suitable tool t^* from the toolset T :

$$t^* = \arg \max_{t \in T} P(t|q, C_{retrieved}) \quad (5)$$

After obtaining the tool analysis result R_{tool} , the final diagnosis is made using the combined information $I_{final} = C_{retrieved} \oplus R_{tool}$.

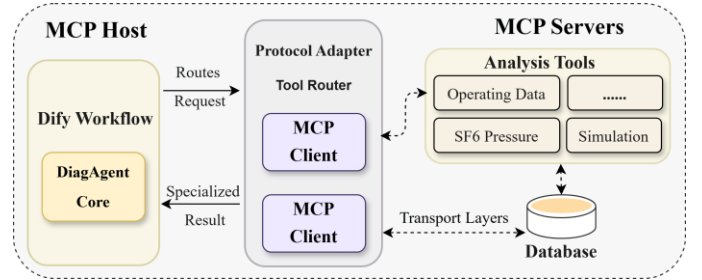


Figure 2. Architecture for MCP-based tool invocation caption

C. Diagnostic Report Generation

Using a template engine (DocxTpl) [12], the LLM's final diagnosis is structured into a standardized report format, including sections for observed phenomena, analysis, conclusion, and recommendations, ensuring clarity and actionability for engineers.

Report generation can be viewed as a conditional text generation task. Given the final information I_{final} and the template T , the probability of generating the report R is shown in (6):

$$P(R|q) = \prod_{i=1}^L P(r_i | r_{<i}, T, I_{final}, q) \quad (6)$$

The model fills in key information based on the template structure to generate the final report.

IV. EXPERIMENTS AND EVALUATION

A. Experimental Setup

The evaluation utilizes a dataset of 150 real-world historical fault cases of power bushings. Performance is assessed using two primary metrics: diagnostic accuracy, determined by direct comparison with domain expert conclusions, and report quality score, where three domain experts rate each report on a 1-5 scale across accuracy, completeness, and clarity.

This work benchmarks the proposed method against two Qwen-30B configurations: the vanilla Qwen-30B with zero-shot prompting, and the Qwen-30B augmented with a naive RAG pipeline using single-step vector retrieval.

B. Results and Analysis

The main results are summarized in Table I. DiagAgent achieves a significant performance improvement, demonstrating the effectiveness of the integrated framework.

TABLE I. MAIN EXPERIMENTAL RESULTS

Method	Accuracy	Report Score
Qwen-30B (Zero-shot)	65.3%	2.8
Qwen-30B + Naive RAG	80.7%	3.5
DiagAgent (Ours)	91.3%^a	4.4^a

a. Bold font indicates the state-of-the-art results.

The results in Table I demonstrate the effectiveness of the integrated DiagAgent framework. The vanilla Qwen-30B model (65.3% accuracy) highlights the insufficient domain-specific knowledge and analytical capability of a general-purpose LLM for complex diagnosis. While augmenting it with a naive RAG pipeline (80.7% accuracy) significantly improves performance by grounding the model in external knowledge, it remains limited by single-step retrieval and the lack of deep analytical tools. In contrast, DiagAgent achieves the best performance (91.3% accuracy, 4.4 report score) by synergizing multi-stage hybrid retrieval (for precise, multi-perspective knowledge fetching) with on-demand MCP tool invocation (for specialized numerical and signal analysis). This validates our core design premise: that combining context-aware knowledge retrieval with executable analytical functions is essential for achieving reliable, high-quality diagnostic outcomes that meet expert standards.

To validate the contribution of each key component, we conducted ablation studies as summarized in Table II.

TABLE II. ABLATION RESULTS

Method	+ MCP	+ RAG	Accuracy
DiagAgent	✓		84.5%
		✓	85.1%
	✓	✓	91.3%^a

a Bold font indicates the state-of-the-art results.

In the table, +MCP denotes the integration of MCP-based tool invocation for proprietary small models into the base framework, while +RAG denotes the addition of the Multi-Stage RAG module. The results confirm that both modules are indispensable. The RAG module provides reliable, context-aware knowledge, a recognized enhancement over standalone LLMs [7], while the MCP module enables the deep analytical computations that typical RAG-enhanced QA systems lack [8]. Their synergy in DiagAgent is key to achieving superior performance, as removing either component leads to a significant accuracy drop.

C. Case Study

Taking the bushing fault case of "elevated SF₆ gas leakage" as an example, a diagnostic report was generated using

DiagAgent and tested on the Dify platform, with the workflow illustrated in Figure 3.

DiagAgent receives user input, undergoes processes including RAG, template comprehension, and MCP tool invocation, and ultimately produces a diagnostic report.

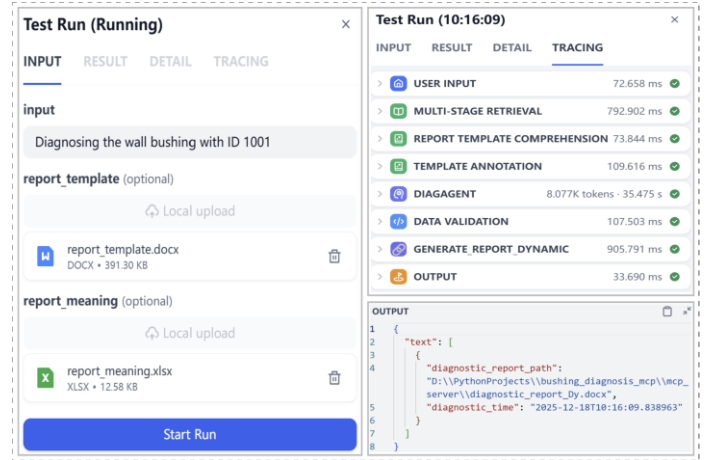


Figure 3. Diagnostic Testing Workflow

The generated diagnostic report is shown in Figure 4. The report clearly presents the abnormal parameter analysis, risk assessment, and specific maintenance recommendations, demonstrating the framework's capability to transform raw data into actionable insights for critical infrastructure.

V. CONCLUSIONS

This paper presented DiagAgent, a practical and effective framework for power equipment fault diagnosis. By integrating RAG with MCP-based tool calling within an LLM agent, DiagAgent successfully mitigates key limitations of standalone LLMs, combining reliable knowledge access with deep analytical power. On real-world bushing fault cases, DiagAgent achieved a diagnostic accuracy of 91.3% and a report quality score of 4.4/5. It provides a scalable blueprint for building trustworthy AI assistants in industrial settings.

However, the framework still has limitations. First, its diagnostic accuracy is inherently contingent on the quality and coverage of the underlying knowledge base; consequently, novel or insufficiently documented faults may not be handled effectively. Second, the latency introduced by the multi-stage retrieval and tool invocation pipeline may not satisfy the requirements for millisecond-level real-time fault response in certain critical scenarios.

These limitations define clear directions for future work. We will focus on multi-modal data integration to enhance perceptual capabilities and on implementing an automated knowledge base update pipeline to continuously integrate new field cases and research findings, thereby extending the system's applicability and robustness.

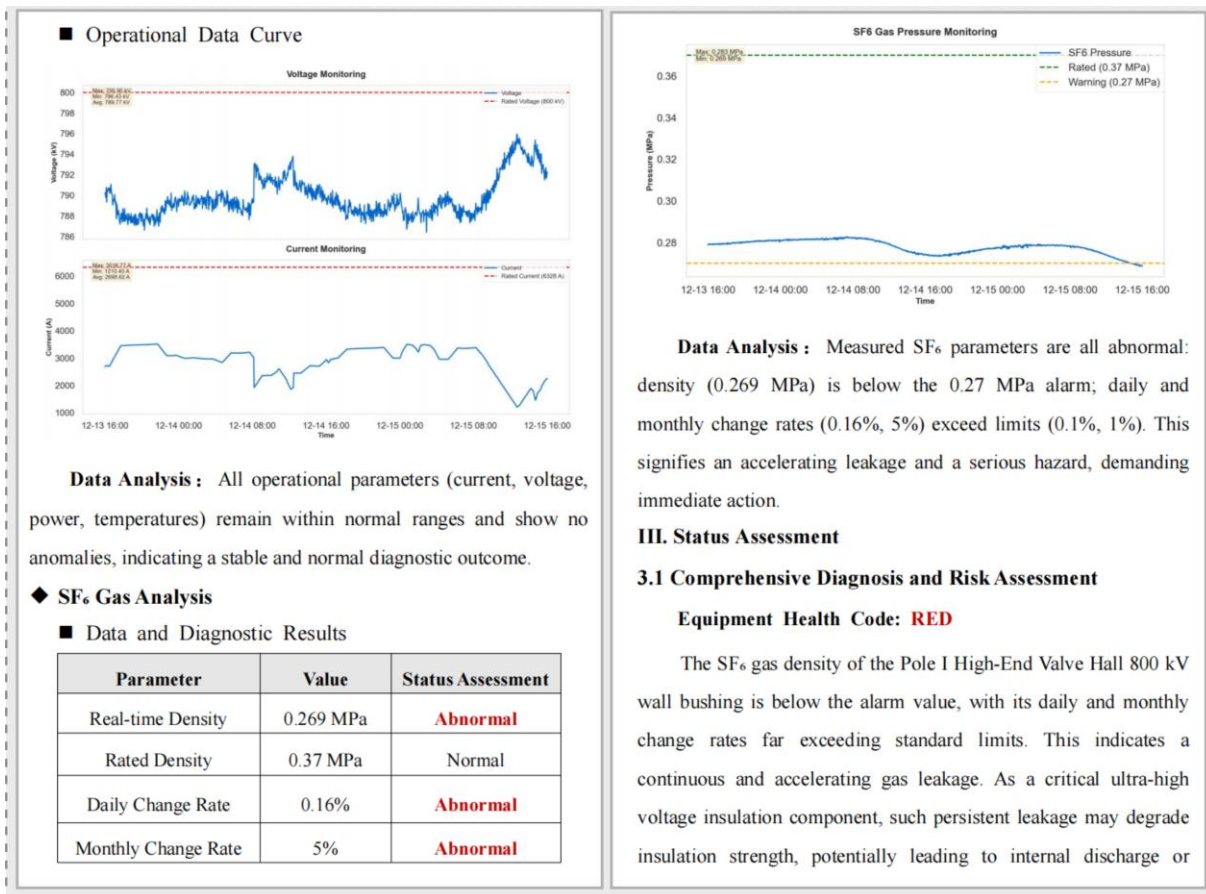


Figure 4. Case study report: SF₆ leakage diagnosis

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